SESCO CEMENT CORPORATION

Natural Pozzolans & LC3 Comprehensive Market Dossier

Volcanic Ash, Pumice, Calcined Clay & Metakaolin
SCM Markets, LC3 Emergence & Supply Outlook

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NATURAL POZZOLANS COMPREHENSIVE DOSSIER

Prepared: February 18, 2026 | **Version:** 1.0 | **Focus:** SCM Markets, LC3 Emergence & Supply Outlook as Fly Ash/GGBFS Decline

EXECUTIVE SUMMARY: Natural pozzolans are emerging as the critical replacement SCM as both fly ash and GGBFS face structural supply declines. US natural pozzolan shipments to concrete reached 2.5 million tons in 2024 (NPA data), with capacity projected to reach 3.5–4.0 Mt by 2028. Unlike fly ash (tied to coal) and GGBFS (tied to blast furnaces), natural pozzolans are independent mineral resources with large identified US reserves (>25 Mt pumice alone; Kirkland Mine 39 Mt). The DOE has invested $1.6 billion in cement decarbonization, with 3 of 6 funded projects targeting LC3 (Limestone Calcined Clay Cement) — which can cut CO2 by 40% using widely available low-grade clays. Fly ash prices have risen to $118/Mt (from $40–60 historically), making natural pozzolans at ~$115/Mt delivered increasingly cost-competitive. Major cement companies are moving: CRH/Ash Grove acquired Geofortis pozzolan operations (Sep 2024); Eco Material has >4 Mt/yr SCM capacity across 9 plants. This is the growth sector of the SCM market.

2.5 Mt
US Nat. Pozzolan Shipments (2024)
3.5-4.0 Mt capacity by 2028

$1.6B
DOE Cement Decarb Investment
3 of 6 projects = LC3

~$115
Nat. Pozzolan $/Mt Delivered
Now competitive with fly ash $118

40%
LC3 CO2 Reduction vs OPC
Using low-grade clay + limestone

Executive Briefing — Strategic Implications for SESCO Cement

Natural pozzolans represent the growth sector of the SCM market and a strategic diversification opportunity as both fly ash and GGBFS face structural supply declines. This briefing highlights the five most critical takeaways:

- **Natural pozzolans are the only growing domestic SCM.** US shipments reached 2.5 Mt (2024) with capacity projected to reach 3.5–4.0 Mt by 2028. Unlike fly ash (declining with coal) and GGBFS (declining with blast furnaces), natural pozzolans are independent mineral resources with large US reserves and no industrial transition risk.
- **LC3 is the long-term cement game-changer.** Limestone Calcined Clay Cement can cut CO2 by 40% using widely available low-grade clays. The DOE has invested $1.6B in cement decarbonization, with 3 of 6 funded projects targeting LC3. **SESCO opportunity:** Understanding LC3 technology positions SESCO ahead of the curve as US adoption scales from pilot to commercial (2026–2030).
- **Price parity with fly ash has arrived.** Natural pozzolans at ~$115/Mt delivered are now competitive with fly ash at $118–123/Mt. As fly ash prices continue rising (supply declining), natural pozzolans become the clear cost-effective alternative. **SESCO opportunity:** Blending imported cement with domestic natural pozzolans creates a differentiated, cost-competitive, low-carbon product.
- **Major cement companies are moving.** CRH/Ash Grove acquired Geofortis pozzolan operations (Sep 2024). Eco Material Technologies has >4 Mt/yr SCM capacity across 9 plants. Boral, Hess Pumice, and CR Minerals are expanding. First-mover advantage in pozzolan sourcing is narrowing.
- **Import potential from volcanic regions.** Greece, Italy, Turkey, and Central America have abundant volcanic pozzolan deposits. Turkey is already a SESCO cement supply partner. **SESCO opportunity:** Co-sourcing natural pozzolans with cement from Turkey or Mediterranean origins leverages existing trade lanes and vessel logistics.

1. TYPES OF NATURAL POZZOLANS

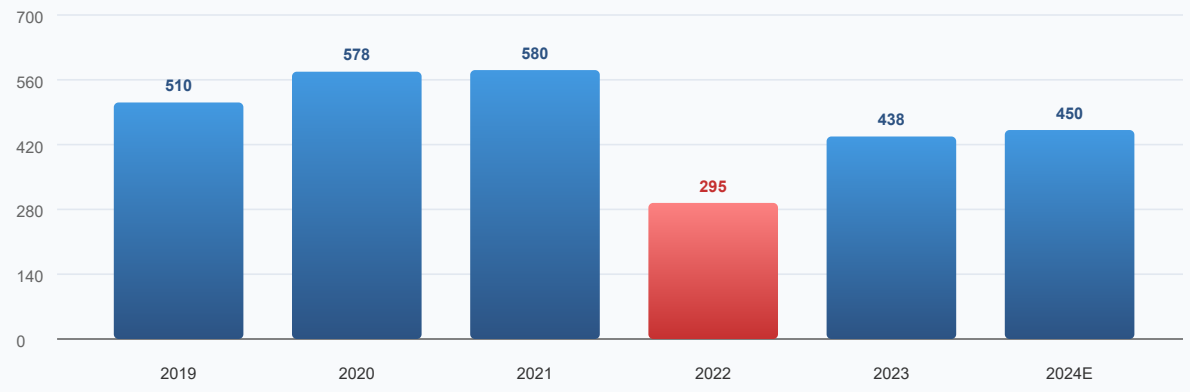
Type	Source / Origin	Key Properties	Primary Uses	US Occurrence
Volcanic Ash / Tuff	Volcanic eruption fragments; tuff = consolidated ash	Glassy, amorphous SiO2 52–66%; glass content 50–97%	SCM in concrete (15–40% replacement)	Pacific NW, ID, NM, AZ, NV

Pumice / Pumicite	Vesicular volcanic glass; pumicite = finer fraction	Lightweight (<35 lb/ft³); $SiO_2+Al_2O_3+Fe_2O_3$ >70%; high surface area	SCM, lightweight aggregate, abrasives, horticulture	ID (Hess), NM (CR Minerals), CA, OR, KS
Calcined Clay	Kaolinitic/illitic clay thermally activated at 600–850°C	Reactive amorphous aluminosilicate; low-grade clays (40% kaolin) viable	LC3 cement (30% replacement); SCM in concrete	GA, TN, SE kaolin belt
Metakaolin	High-purity kaolin clay calcined at 650–800°C	$SiO_2+Al_2O_3+Fe_2O_3$ >90%; avg particle 1.4 µm; white; highly reactive	High-performance concrete SCM (5–20% replacement); fiber cement	GA (Burgess Pigment)
Diatomaceous Earth	Siliceous sedimentary rock (fossilized diatoms)	80–90% void; BET 100–200 m²/g; controls ASR; higher water demand	SCM (5–20%), filtration, insulation, absorbents	CA, OR, NV, WA
Perlite	Volcanic glass with 2–5% water; expands 4–20x when heated	Extremely lightweight; fire resistant; excellent insulator	Building products (44%), horticulture (21%), fillers, filtration	Western states; US market $370M (2024)
Calcined Shale	Shale/siltstone calcined in rotary kilns at 600–900°C	Activated pozzolanic reactivity after thermal treatment	SCM in concrete; road stabilization	TX, Midwest
Rice Husk Ash	Controlled combustion of rice husks (agricultural byproduct)	SiO_2 up to 80%; high activity when amorphous; ~20 Mt RHA/yr globally	SCM (10–20% replacement); market $600M (2024) → $1.2B by 2033	AR, CA, LA (rice belt)
Trass	Volcanic tuff from Rhine region (Germany/Netherlands)	Traditional European pozzolan; centuries of proven use	Cement, mortar, historic restoration	N/A (European source)

ASTM C618 CLASS N: All natural pozzolans used in concrete must meet ASTM C618 Class N requirements: $SiO_2+Al_2O_3+Fe_2O_3 \geq 70\%$, Strength Activity Index ≥ 75% at 7 and 28 days, fineness ≤ 34% retained on 45 µm sieve, water requirement ≤ 115% of control.

2. US PRODUCTION

2.1 Pumice & Pumicite (USGS Data)



Source: USGS Mineral Commodity Summaries 2024-2025. 10 operations in 5 states (CA, ID, KS, NM, OR). US identified pumice resources: >25 million tons.

2.2 Natural Pozzolan Shipments to Concrete Market (NPA Data)

Year	NPA Member Shipments (Mt)	Capacity Outlook
2024	2.5	Growing rapidly
2028 (projected)	—	3.5–4.0 Mt capacity

2.3 Key Mining Operations

Operation	Location	Operator	Reserves/Capacity	Status
Hess Pumice Deposit	Malad City, ID	Hess Pumice Products	Purest commercial white pumice globally; 60+ yrs	Active
CR Minerals	Santa Fe, NM	CR Minerals	~18% revenue share; vertically integrated	Active
Kirkland Mine	Kirkland, AZ	Kirkland Mining / Eco Material	39 Mt reserves; 500K t/yr capacity; 40-yr life	Active (opened 2022)
Geofortis/Ash Grove	Tooele, UT	CRH / Ash Grove	Acquired Sep 2024; approved in CA, CO, NV, UT	Active

Hazen Pozzolan Project	Churchill Co., NV	Sunrise Resources	Glassy pumice; 9 km from UP rail siding; ASTM C618 compliant	Development
Nevada Pozzolan	Nevada	Nevada Pozzolan Ltd	Volcanic tuff deposit up to 980 ft thick	Development
PozzSource Deposits	Lassen Co., CA	PozzSource LLC (fka Geofortis)	R&D lab in Reno; deposit development	Development

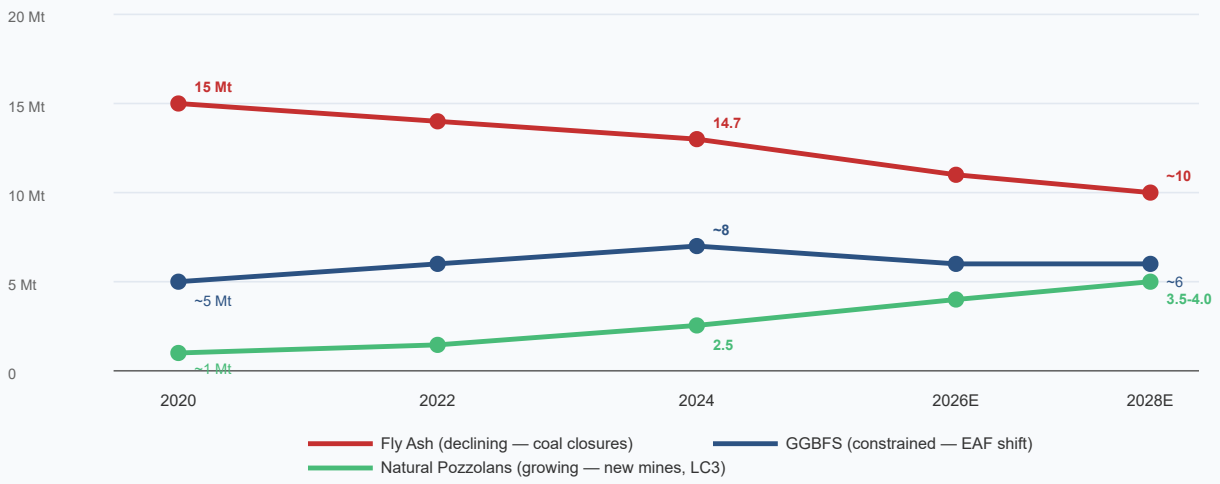
3. MARKET SIZE & FORECASTS

3.1 Natural Pozzolan Markets

Market Segment	2024 Value	Forecast	CAGR
Global Pozzolan Market (all types)	$3.5–4.6 billion	$6.2B by 2032	7.1%
Global Natural Pozzolan Market	$865M–$1.76B	$1.25–3.84B by 2031–33	5.5–9.5%
Global Metakaolin Market	$151–809M	$210M–$1.07B by 2031–33	4.9–5.5%
US Perlite Market	$370 million	$632M by 2033	6.1%
Global Rice Husk Ash Market	$600 million	$1.2B by 2033	8.5%
Global SCM Market (total)	$91.1 billion	$119.9B by 2033	3.1%
McKinsey SCM Revenue Projection	$15–30 billion	$40–60B by 2035	—

4. THE SCM SUPPLY CRISIS — WHY NATURAL POZZOLANS MATTER NOW

US SCM Supply Landscape — Diverging Trajectories



Conceptual projection based on ACAA, USGS, NPA data. Fly ash decline driven by coal plant closures; GGBFS constrained by BF-to-EAF transition; natural pozzolans growing from new mines and LC3 development.

4.1 US SCM Consumption (2024)

SCM Type	2024 Volume (Mt)	Supply Trend	Price ($/Mt)	Supply Independence
Coal Fly Ash (live)	14.7	Declining (coal closures)	$118	Tied to coal power
Slag Cement (GGBFS)	~8.0	Constrained (EAF shift)	$100–140+	Tied to blast furnaces
Natural Pozzolans	2.5	Growing rapidly	~$115	Independent mineral resource
Silica Fume	<1.0	Stable (niche)	$400–600+	Tied to silicon/ferrosilicon
Total SCM to Concrete	**>20**	**—**	**—**	**—**

DUAL SCM CRISIS + SOLUTION: US fly ash supply is declining because coal plants are closing (-64% coal consumption since 2007 peak). GGBFS supply is constrained because blast furnaces are being replaced by EAFs (~70% of US steelmaking). Combined, the two historically dominant SCMs face irreversible structural decline. Natural pozzolans are the only SCM category with **expanding supply from independent mineral reserves** — not dependent on any other industry's production decisions. This is why major cement companies (CRH, Eco Material, National Cement) are investing aggressively.

5. PRICING

$163
Portland Cement (US, 2025)
5.4% CAGR (2020-25)

$171
Metakaolin (US, Q3 2025)
Premium calcined product

$118
Fly Ash (US, Q3 2025)
Up from $40-60 historical

~$115
Natural Pozzolan (delivered)
Now competitive with fly ash

~$42
Pumice (FOB plant)
$19M / 450K tons

5.1 Price Dynamics

Material	2020 Price	2025 Price	Change	Driver
Portland Cement	~$130/Mt	$163/Mt	+25%	Input costs, demand
Fly Ash	~$40–60/Mt	$118/Mt	+100–195%	Supply scarcity (coal closures)
Natural Pozzolan	~$90–100/Mt	~$115/Mt	+15–28%	Modest; supply expandable
Metakaolin	~$140/Mt	$171/Mt	+22%	Calcination energy costs

PRICE CONVERGENCE: The critical shift is that fly ash has risen from $40–60/Mt to $118/Mt while natural pozzolans remain at ~$115/Mt. For the first time, natural pozzolans are **cost-competitive with fly ash** on a delivered basis. This price crossover is the fundamental driver of natural pozzolan market growth — and unlike fly ash, natural pozzolan supply can expand to meet demand.

6. IMPORTS & EXPORTS

6.1 US Pumice Trade

Category	Volume (2024)	Key Sources	Notes
Imports	~55,000 Mt	Greece (91%), Iceland (8%)	Crude pumice for eastern/Gulf Coast markets

Exports	~12,500 Mt	Various	Processed pumice products
Net Position	Net Importer	—	Imports serve markets far from Western US deposits

6.2 HTS Codes

HTS Code	Description	Relevance
2513.10.0010	Pumice, crude or in irregular pieces	Primary import code for raw pumice
2513.10.0080	Pumice, except crude or crushed	Processed pumice products
2530.90	Other mineral substances NES	Some volcanic ash/pozzolans classified here

TRADE CONTEXT: Unlike fly ash and GGBFS (where imports are critical supply supplements), natural pozzolan trade volumes are small. The US has abundant domestic reserves (>25 Mt pumice alone) concentrated in Western states. Greek pumice imports serve Eastern/Gulf markets where domestic sources are distant. The key supply chain challenge is **West-to-East distribution** rather than import dependency.

7. MAJOR PRODUCERS & SUPPLIERS

7.1 Natural Pozzolan Producers

Company	Location	Product	Key Details
Hess Pumice Products	Malad City, ID	HessPozz; processed pumice	Largest processed pumice producer globally; 60+ year history; purest white pumice deposit
CR Minerals	Santa Fe, NM	Pumice products; oil & gas applications	~18% revenue share; vertically integrated mining-processing-distribution
Eco Material Technologies	National (45 states)	Fly ash, natural pozzolan, Green Cement	>50% fly ash market share; 9 plants; >4 Mt/yr SCM capacity; 4,000+ customer locations
Kirkland Mining / Eco Material	Kirkland, AZ	Pumiceous tuff natural pozzolan	39 Mt reserves; 500K t/yr; 88 acres; 40-yr mine life; opened 2022
Ash Grove / CRH	Tooele, UT	Natural pozzolan milling & classifying	Acquired Geofortis ops Sep 2024; approved in CA, CO, NV, UT

St. Cloud Mining	NM, AZ, CA, NV, OR	Zeolite & volcanic ash	Largest characterized zeolite reserves in US; ASTM-qualified for pozzolanic cements
Sunrise Resources	Churchill Co., NV	Hazen Pozzolan Project	Glassy pumice; near UP rail siding; ASTM C618 compliant; development stage
Nevada Pozzolan Ltd	Nevada	Volcanic tuff pozzolan	Deposit up to 980 ft thick; development stage
Environmental Minerals Inc.	Las Vegas, NV	ASTM C618 Class N pozzolan	Low-carbon concrete additives; also rare earth minerals

7.2 Metakaolin / Calcined Clay Producers

Company	Location	Product	Key Details
Burgess Pigment	Sandersville, GA	OPTIPOZZ metakaolin	Avg particle 1.4 µm; Class N pozzolan; SE kaolin belt
Advanced Cement Technologies	—	PowerPozz High-Reactivity Metakaolin	High-performance concrete specialist since 1995
Imerys (fka BASF)	Various	MetaMax metakaolin	Used in 432 Park Ave NYC; high-purity kaolin

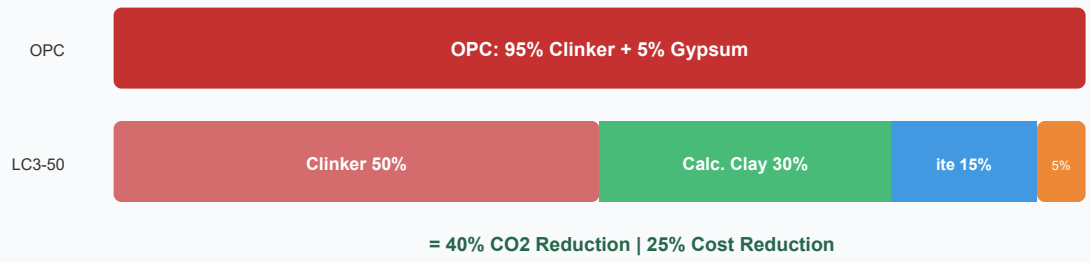
7.3 Cement Companies Investing in Natural Pozzolans / LC3

Company	Project	Investment	Details
National Cement Co. of CA	Lebec, CA — net-zero plant	Up to $500M (DOE)	LC3 + biomass + CCS; prevent 2 Mt CO2/yr
Roanoke Cement	Troutville, VA — LC3 demo	Part of $1.6B DOE	83% carbon intensity reduction using widely available clays
Summit Materials	LC3 demonstration	$215.6M (DOE potential)	Low-carbon calcined clay cement
CRH / Ash Grove	Geofortis pozzolan acquisition	N/D	Acquired Sep 2024; secures natural pozzolan supply
CalPortland	ADVANCEMENT eco-cement	N/D	Western US; eco-friendly cement products

8. LC3 — LIMESTONE CALCINED CLAY CEMENT

8.1 Technology Overview

LC3-50 Cement Composition vs Ordinary Portland Cement



LC3-50 replaces 45% of clinker with calcined clay (30%) + limestone (15%), achieving up to 40% CO2 reduction while maintaining performance. Uses low-grade clays (40% kaolinite sufficient).

8.2 LC3 Key Facts

Attribute	Detail
Clinker content	50% (vs 95% in OPC)
CO2 reduction	Up to 40% vs OPC
Cost reduction	Up to 25% vs OPC
Clay requirement	Low-grade: 40% kaolinite sufficient; illite/montmorillonite also viable
Plant modifications	Minimal — existing cement plant infrastructure compatible
ACEEE impact estimate	Shifting half of US govt cement procurement to LC3 = **7.3 Mt CO2/yr saved**

8.3 Global LC3 Adoption

Region	Status	Key Development
India	**Leading**	BIS released IS 18189:2023 — exclusive LC3 standard
Ghana	**Leading**	CBI Ghana/Accra = largest LC3 plant globally (FLSmidth calciner); 60–70% clinker substitution
Latin America	Growing	Broad traction across region

Europe	Growing	Emerging regulatory support
US	Early	3 DOE-funded demos ($1.6B total); ASTM C595/C618 cover LC3 but limits need expansion

8.4 US LC3 Challenges

- Insufficient calcination infrastructure (few plants have clay calciners)
- ASTM standards allow calcined clay to 40% and limestone to 15% — but not as combined LC3 formulation
- Industry pushing for max 70% for any combination of limestone, pozzolan, and slag
- Some state DOTs extending acceptance testing from 28 to 42 or 56 days
- Historical construction practices favor conventional OPC
- Limited domestic production and suppliers at commercial scale

LC3 OUTLOOK: LC3 is the most scalable long-term alternative to declining fly ash and GGBFS. Raw materials (low-grade clay, limestone) are globally abundant and cheaply available. The $1.6B DOE investment in 6 cement decarbonization projects (3 focused on LC3) signals strong federal commitment. But commercial-scale US deployment remains 3–5 years away. Natural volcanic pozzolans are the near-term bridge SCM while LC3 infrastructure is built.

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9. END USES & APPLICATIONS

9.1 Natural Pozzolan in Cement/Concrete

Application	Typical Replacement Rate	Notes
Portland Pozzolan Cement (PPC / Type IP)	10–25% interground with clinker	ASTM C595 allows up to 40%
Ready-mix concrete SCM	15–40%	General construction
High-performance concrete (metakaolin)	5–20%	Bridges, marine, high-strength
LC3 cement (calcined clay)	30% clay + 15% limestone	Combined 45% clinker replacement
Mass concrete	20–35%	Lower heat of hydration
Oil well cementing	Variable	Especially zeolite-based pozzolans
Soil stabilization	Variable	Geotechnical applications

9.2 Pumice End Uses (Non-Cement)

Application	Description
Lightweight aggregate	Concrete blocks, structural lightweight concrete
Abrasives	Hand soaps, cleaning compounds, polishing
Horticulture	Soil amendment, growing media
Filtration	Water treatment filter media
Laundry stone washing	Textile finishing

10. PHYSICAL PROPERTIES & SPECIFICATIONS

10.1 Chemical Composition (%)

Oxide	Volcanic Ash/Pumice	Metakaolin	Diatomaceous Earth	Fly Ash (Class F)	Portland Cement
SiO_2	52–75	50–55	70–90	45–55	19–23
Al_2O_3	12–18	35–45	2–10	20–30	4–7
Fe_2O_3	1–5	0.5–2.0	1–5	5–15	2–4
CaO	1–5	0.1–0.3	0.5–2	2–8	60–67
$SiO_2+Al_2O_3+Fe_2O_3$	**>70%**	**>90%**	**>75%**	**>70%**	**—**

10.2 ASTM C618 Class N Requirements

Property	Requirement	Typical Values
$SiO_2 + Al_2O_3 + Fe_2O_3$	≥ 70.0%	75–95%
SO_3	≤ 4.0%	<1%
Moisture content	≤ 3.0%	<1%
Loss on ignition	≤ 10.0%	2–6%
Fineness (retained 45 µm)	≤ 34%	5–30%
Strength Activity Index (7d)	≥ 75%	75–90%

Strength Activity Index (28d)	≥ 75%	85–110%
Water requirement	≤ 115% of control	95–115%
Autoclave expansion	≤ 0.8%	<0.1%

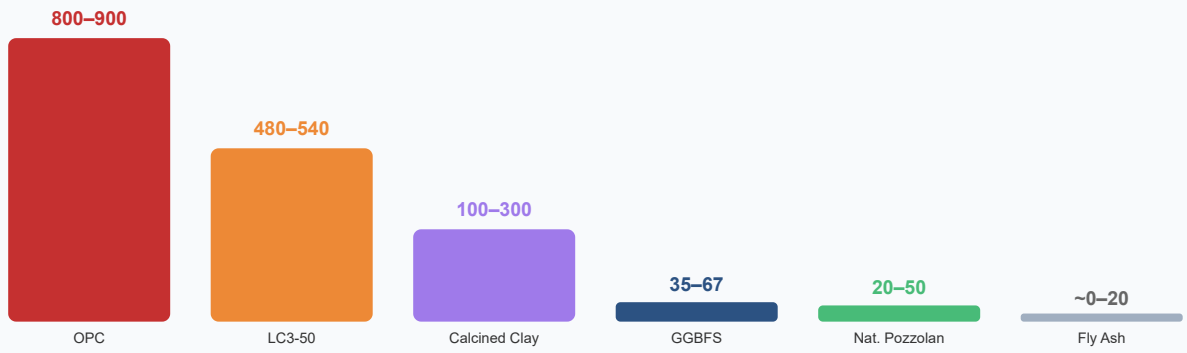
10.3 Replacement Rates by Pozzolan Type

Pozzolan Type	Typical Range	Optimal	Notes
Pumice / volcanic ash	15–40%	20–30%	General concrete; most common
Metakaolin	5–20%	10–15%	High-performance; small dosage, high reactivity
Calcined clay (LC3)	20–40%	30%	Combined with 15% limestone
Diatomaceous earth	5–20%	10–15%	Higher water demand
Rice husk ash	10–20%	10–15%	Fineness dependent

11. CARBON & SUSTAINABILITY

11.1 Embodied CO2 Comparison

Embodied CO2 by Cementitious Material (kg CO2 / tonne)



Natural pozzolans: 95–97% lower CO2 than OPC (mining + grinding only; no calcination)

Sources: PCA, CSMA, SLR Consulting, various EPDs. Natural pozzolan CO2 is mining + crushing + grinding only. Calcined clay/metakaolin includes kiln energy. Fly ash and GGBFS allocated as industrial by-products.

11.2 Regulatory Drivers

Regulation	Requirement	Natural Pozzolan Impact
GSA Buy Clean (IRA)	EPDs required; GWP limits for concrete	**Strong driver**
LEED v5	Mandatory embodied carbon prerequisite	**Strong driver**
Buy Clean CA/CO/NY/OR	State EPD and carbon limits for public works	**Growing driver**
DOE Industrial Demos	$1.6B for cement decarb; 3 LC3 projects	**Direct investment**
EPA EPD Program	Funding for EPD creation; NPA/PCA EPD Upgrade (Jul 2024)	**Enabler**

DURABILITY BONUS: Portland Pozzolan Cement (PPC) with natural pozzolans can deliver up to **5x longer life expectancy** vs conventional OPC (350 years vs 70 years potential). The refined microstructure, reduced permeability, and resistance to sulfate attack and ASR make pozzolanic concrete ideal for infrastructure with 75–100+ year design lives — exactly what IIJA-funded projects require.

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12. GEOGRAPHIC SOURCES & LOGISTICS

12.1 US Volcanic Pozzolan Deposits

Region	States	Geology	Active Operations
Snake River Plain	ID	Large pumice & volcanic tuff deposits	Hess Pumice (Malad City)
Rio Grande Rift	NM	Volcanic ash & pumice near Santa Fe	CR Minerals, St. Cloud Mining
Basin & Range	NV, UT, AZ	Rhyolitic tuff; some deposits up to 980 ft thick	Kirkland Mining (AZ); Nevada Pozzolan; Ash Grove (UT); Sunrise Resources (NV)
Cascade Range	OR, WA, CA	Extensive volcanic ash from Mt. Mazama eruption	Active pumice mining in OR; PozzSource (CA)
Great Plains	KS	Historic Pearlette ash beds from NV/UT sources	Active mining
SE Kaolin Belt	GA, SC, NC, TN	Kaolinite deposits for metakaolin/calcined clay	Burgess Pigment (GA)

12.2 Supply Chain & Logistics

Stage	Process	Notes
Mining	Shallow open-pit; low stripping ratios; minimal blasting	Soft material; 40-yr mine lives typical
Drying	Remove hygroscopic moisture	Required for effective grinding
Grinding/Classifying	Ground to 6–11 μm mean particle size	Finer = higher reactivity; at 6 μm can replace 40% cement
Quality Testing	ASTM C618 Class N compliance	SAI, fineness, chemistry verified
Distribution — Truck	Pneumatic bulk tankers	Primary mode; pumice is lightweight, needs high-cube trailers
Distribution — Rail	Covered hopper cars	Key for long-distance; Hazen NV project near UP rail siding
Storage	Enclosed silos (dry storage required)	Similar to cement/fly ash handling; lower bulk density = more volume needed

GEOGRAPHIC CHALLENGE: Natural pozzolan deposits are concentrated in the **Western US** (ID, NM, AZ, NV, OR, UT, CA), while the largest concrete markets are in the **East and South** (TX, FL, GA, OH, PA, NY). Rail is critical for bridging this gap. Greek pumice imports serve eastern/Gulf Coast markets where domestic Western pozzolans face high distribution costs. The West-to-East logistics cost is the primary constraint on natural pozzolan market growth.

13. REGULATORY & STANDARDS

13.1 Key Standards

Standard	Scope	Key Provision
ASTM C618-25a	Coal Ash & Natural Pozzolans	Class N: natural pozzolans; $SiO_2+Al_2O_3+Fe_2O_3$ ≥70%; SAI ≥75%
ASTM C595/C595M-24	Blended Hydraulic Cements	Type IP: up to 40% pozzolan; Type IT: ternary blends
ASTM C1157	Performance Hydraulic Cement	Any SCM proportion if performance criteria met
ASTM C1240	Silica Fume	Max 10% of cementitious; pozzolans max 25%

ASTM C1697	Blended SCMs	Covers SCM combinations
ASTM C1945	Natural Pozzolan Grade	Newer standard for NPA-grade products

13.2 State DOT Acceptance

State	Specification	Key Provision
Texas (TxDOT)	DMS-4635	Comprehensive nat. pozzolan spec; CCRL lab inspection; approved materials list
New York	Section 501	All classes except F allow pozzolan; up to 20% substitution
Washington	M 41-10 Div. 9	Class N natural pozzolan: max 35% in concrete
California (Caltrans)	Auth. Materials List	NP200C authorized for concrete use
Ohio DOT	CMS 499	NP100W on Qualified Products List
Wisconsin	2025 Std Specs	Natural pozzolan recognized as SCM
Florida DOT	Proposed changes	Identifying natural pozzolan as SCM; reflecting ASTM changes
Minnesota DOT	NRRA Rigid Team	Reducing cement content with pozzolanic materials research
CO, NV, UT	Various	Geofortis/Ash Grove product approved

EXTENDED TESTING: Multiple state DOTs are extending acceptance testing from 28 days to **42 or 56 days** to accommodate the slower pozzolanic reaction of natural pozzolans. This is critical: natural pozzolans develop strength more slowly than OPC but achieve equal or superior long-term strength and durability.

14. SUPPLY OUTLOOK & TRENDS

Natural Pozzolans — 5-Year Outlook (2025–2030)

Factor	Trend	Impact on Supply	Impact on Price
Fly Ash Decline	Coal closures; CCP 66.7 → 63.6 Mt (2023–24); no new plants	**DEMAND SHIFT to nat. pozz**	**UP (fly ash scarcity)**

GGBFS Constraint	EAF shift (70%); BF closures; 22% of cement producers report shortages	**DEMAND SHIFT to nat. pozz**	**UP (GGBFS scarcity)**
New Pozzolan Mines	Kirkland (39 Mt), Hazen (NV), Nevada Pozzolan, PozzSource (CA)	**SUPPLY EXPANDING**	**DOWN (competition)**
Major Company Entry	CRH/Ash Grove acquires Geofortis; Eco Material 4+ Mt/yr capacity	**SUPPLY EXPANDING**	NEUTRAL (quality premium)
DOE LC3 Investment	$1.6B; 3 of 6 projects target LC3; 3–5 yr deployment	**LONG-TERM SUPPLY**	**DOWN (long-term)**
Buy Clean / LEED v5	Embodied carbon mandates; 14K+ new EPDs; federal & state	N/A (demand driver)	**UP (demand pull)**
State DOT Acceptance	TX, NY, WA, CA, OH, WI, FL all accepting or proposing nat. pozz specs	**MARKET EXPANSION**	NEUTRAL
Price Convergence	Fly ash $118/Mt vs nat. pozz ~$115/Mt; first time competitive	**DEMAND SHIFT accelerating**	STABLE

14.1 Natural Pozzolans: Advantages vs Limitations

Advantages

- Independent mineral resource — not tied to coal or steel
- Consistent quality (geological deposit vs variable industrial byproduct)
- Large identified US reserves (>25 Mt pumice; 39 Mt Kirkland; massive NV deposits)
- Simple processing: mine, dry, grind, classify
- Now cost-competitive with fly ash (~$115 vs $118)
- ASTM C618 Class N compliant
- Superior long-term durability (up to 5x OPC lifespan)
- 95–97% lower embodied CO2 than OPC

Limitations

- Geographic concentration in Western US; high distribution costs to East
- Slower early strength development (7-day) vs OPC
- Higher water demand for some types (diatomaceous earth)
- Quality varies by deposit; not all volcanic materials qualify
- Lighter bulk density = more storage volume needed
- Calcined clay/metakaolin requires energy for kiln processing
- Market awareness still growing; some specifiers unfamiliar
- State DOT acceptance still expanding (not yet universal)

NET ASSESSMENT: Natural pozzolans are positioned for strong growth as the **only expanding-supply SCM category** in the US. The convergence of fly ash price increases ($118/Mt), GGBFS supply constraints, new mine openings (Kirkland 39 Mt, Hazen, Nevada deposits), major company investment (CRH, Eco Material), federal LC3 funding ($1.6B DOE), and carbon regulation (Buy Clean, LEED v5) creates a multi-year growth runway. NPA projects capacity growing from 2.5 Mt (2024) to 3.5–4.0 Mt by 2028 — and LC3 commercial deployment starting 2028–2030 could multiply calcined clay demand by an order of magnitude. The primary constraints are Western geographic concentration and the 3–5 year lead time for LC3 infrastructure. Natural volcanic pozzolans serve as the **near-term bridge SCM** while LC3 scales.

15. QUICK REFERENCE — KEY NUMBERS

Metric	Value	Trend
US Pumice Production (2024)	450K tons, $19M	Recovering from 2022 dip
US Nat. Pozzolan Shipments to Concrete (2024)	2.5 Mt (NPA data)	Growing rapidly
Projected NP Capacity (2028)	3.5–4.0 Mt	+40–60% growth
Total US SCM Shipments (2024)	>20 Mt	Fly ash + GGBFS + NP + SF
US Fly Ash to Concrete (2024)	14.7 Mt	Declining (coal closures)
US GGBFS to Concrete (2024)	~8.0 Mt	Constrained (EAF shift)
Natural Pozzolan Price (delivered)	~$115/Mt	Now competitive with fly ash
Fly Ash Price (US, Q3 2025)	$118/Mt	Up from $40–60 historical
Metakaolin Price (US, Q3 2025)	$171/Mt	Premium calcined product
Portland Cement Price (US, 2025)	$163/Mt	5.4% CAGR (2020–25)
Kirkland Mine Reserves	39 million tons	500K t/yr; 40-yr mine life
US Pumice Reserves (identified)	>25 million tons	Western US
DOE LC3 Investment	$1.6B (6 projects, 3 LC3)	Major federal commitment
LC3 CO2 Reduction vs OPC	Up to 40%	Using low-grade clay + limestone
Natural Pozzolan Embodied CO2	20–50 kg CO2/t	95–97% lower than OPC
Global Natural Pozzolan Market	$865M–$1.76B (2024)	$1.25–3.84B by 2031–33
McKinsey SCM Market (2035)	$40–60 billion	From $15–30B today

World Pumice Production (2024)	~18 Mt	Turkey #1, Greece #2
Global Metakaolin Production (2025)	~1.55 Mt	Asia-Pacific 40% share
Eco Material SCM Capacity	>4 Mt/yr (9 plants, 45 states)	Expanding
ACEEE LC3 Impact Estimate	7.3 Mt CO2/yr saved	If half of US govt procurement shifts

SOURCES

Government: USGS Mineral Commodity Summaries 2024-2025 (Pumice, Perlite); DOE Office of Clean Energy Demonstrations — Industrial Demonstrations Program; EPA Comprehensive Procurement Guidelines; FHWA Natural Pozzolan SCMs Research; US Army ERDC TR-11-46; ACEEE Research Report i2401
Standards: ASTM C618-25a (Coal Ash & Natural Pozzolans); ASTM C595/C595M-24 (Blended Cement); ASTM C1157; ASTM C1240; ASTM C1697; ASTM C1945
Industry Organizations: National Pozzolan Association (NPA); Slag Cement Association; ACAA (2024 Production & Use Survey); Portland Cement Association (PCA)
Companies: Hess Pumice Products; CR Minerals; Eco Material Technologies; Kirkland Mining; Ash Grove/CRH (Geofortis acquisition); PozzSource LLC; Sunrise Resources; Nevada Pozzolan Ltd; St. Cloud Mining; Burgess Pigment (OPTIPOZZ); Advanced Cement Technologies; CalPortland
LC3: LC3.ch (official); RMI — Business Case for LC3 (Dec 2024); FLSmidth (Ghana clay calciner); BIS IS 18189:2023 (India LC3 standard)
Market Research: Technavio (Pozzolan Market); Intel Market Research (Natural Pozzolans); Verified Market Reports; IMARC Group (Perlite, Fly Ash Pricing); Cognitive Market Research (Metakaolin); ChemAnalyst (Metakaolin Pricing); McKinsey (Future of SCMs)
Trade Publications: Concrete Products Magazine; Pit & Quarry; Precast.org; Global Cement Magazine
State DOTs: TxDOT DMS-4635; NYSDOT Section 501; WSDOT M 41-10; Caltrans Authorized Materials; ODOT CMS 499; WisDOT 2025 Standard Specifications